

3-g Tower of Hanoi in Prolog

PROGRAM:

```
hanoi(1, Source, Dest, _) :-  
    format('Move disk from ~w to ~w~n', [Source, Dest]).  
  
hanoi(N, Source, Dest, Aux) :-  
    N > 1,  
    M is N - 1,  
    hanoi(M, Source, Aux, Dest),  
    hanoi(1, Source, Dest, Aux),  
    hanoi(M, Aux, Dest, Source).  
  
run_hanoi :-  
    write('Enter number of disks: '), read(N),  
    hanoi(N, left, right, middle).
```

OUTPUT:

```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

28 ?- consult(['c:/Users/kames/OneDrive/Documents/toh.pl']).
true.

29 ?- run_hanoi.
Enter number of disks: 3.
Move disk from left to right
Move disk from left to middle
Move disk from right to middle
Move disk from left to right
Move disk from middle to left
Move disk from middle to right
Move disk from left to right
true.

30 ?- ▲
```